Langmuir
Article
pubs.acs.org/Langmuir
© 2010 American Chemical Society

# A Microtensiometer To Probe the Effect of Radius of Curvature on Surfactant Transport to a Spherical Interface

Nicolas J. Alvarez,[†] Lynn M. Walker,[†] and Shelley L. Anna*[,†,‡]

[†]*Department of Chemical Engineering and* [‡]*Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, Pennsylvania 15213*

*Received May 10, 2010. Revised Manuscript Received June 25, 2010*

Diffusion of surfactant to a spherical interface depends on the radius of curvature of the interface; the smaller the radius of curvature is, the faster the dynamics. This paper presents and validates an experimental apparatus, denoted a "microtensiometer", to study the dependence of surfactant dynamics on radius of curvature. Dynamic surface tension is monitored for a range of bubble radii from 17 to 150 $\mu$m, and the dynamics are compared with those obtained using the classic pendant drop experiment for a nonionic surfactant at the air−water interface. Experiments reveal that dynamic surface tension follows a diffusion-limited scaling, in which radius of curvature is a key parameter. Despite the clear scaling behavior of the experimental equilibration time, the full dynamic curve for an initially clean interface cannot be predicted by a diffusion-limited transport model using the molecular diffusion coefficient and a single isotherm. However, the same model is shown to correctly predict compression−expansion experiments. Aside from elucidation of surfactant transport, this device provides a tool for rapid measurements of interfacial properties using a significantly lower volume of sample than current methods.

## Introduction

The evolution of surface tension is studied to determine the fundamental transport properties (diffusion, adsorption, and desorption) that dictate how surfactants populate fluid−fluid interfaces. The coefficients of these transport properties are fundamental parameters of the surfactant solution and do not depend on the measurement technique or experimental apparatus. The need to determine these coefficients with certainty is of great importance to many active fields of research, namely surfactant chemistry, interfacial science, and fluid dynamics. In addition, surfactant transport is important in applications involving emulsion stability,[1] drop spreading,[2] drop impact,[3] and drop breakup.[4,5] The ability to quantitatively measure these fundamental parameters will yield a means to catalogue and organize surfactants into classes with particular properties. Furthermore, a better understanding of how changes in surfactant molecular structure influence the kinetic coefficients of adsorption and desorption will assist in the advancement and design of efficient surfactants.

The transport of surfactant to an initially clean spherical or planar interface is understood to follow three simultaneous transport processes.[6−10] Surfactant near the interface undergoes adsorption and desorption from the bulk to the interface and vice versa due to an entropic driving force. The interfacial tension decreases with adsorption of surfactant to the interface. In the absence of convective transport, the surfactant that is depleted near the interface is repopulated with surfactant from the bulk diffusing along the established concentration gradient. The surfactant molecule may also undergo reorientation at the interface. The surface tension stops changing when the interface reaches equilibrium with the bulk and the concentration of molecules on the interface, $\Gamma$, is constant (i.e., $\Gamma = \Gamma_{eq}$, the equilibrium surface concentration). The process is illustrated in Figure 1. Note that although these processes are occurring simultaneously, the relative importance of each process will depend on its individual time scale compared with all other time scales.

When the interface is curved, as for a spherical bubble, the characteristic time scale for molecular diffusion depends on the interface curvature. The ratio of the bubble surface area to the volume surrounding the bubble decreases with decreasing radius. In other words, there are more molecules per unit area available for adsorption near a spherical interface than for a planar interface. This increases the rate of mass transfer to the sphere (i.e., reduces the time scale for diffusion). Scaling arguments reveal that the correct time scale governing diffusion to a spherical interface is a nonlinear function of the spherical depletion depth, $h_s$, and the planar intrinsic length scale, $h_p = \Gamma_{eq}/C_{bulk}$

$$\tau_{Ds} = \frac{(h_s{}^3 h_p)^{1/2}}{D} \quad (1)$$

*To whom correspondence should be addressed. E-mail: sanna@cmu.edu.

(1) Kim, Y. H.; Koczo, K.; Wasan, D. T. Dynamic film and interfacial tensions in emulsion and foam systems. *J. Colloid Interface Sci.* **1997**, *187* (1), 29−44.
(2) Degennes, P. G. Wetting - Statics and Dynamics. *Rev. Mod. Phys.* **1985**, *57* (3), 827−863.
(3) Yarin, A. L. Drop impact dynamics: splashing, spreading, receding, bouncing... *Annu. Rev. Fluid Mech.* **2006**, *38*, 159−192.
(4) Anna, S. L.; Mayer, H. C. Microscale tipstreaming in a microfluidic flow focusing device. *Phys. Fluids* **2006**, *18* (12), 121512.
(5) Jin, F.; Gupta, N. R.; Stebe, K. J. The detachment of a viscous drop in a viscous solution in the presence of a soluble surfactant. *Phys. Fluids* **2006**, *18* (2), 022103
(6) Pan, R. N.; Green, J.; Maldarelli, C. Theory and experiment on the measurement of kinetic rate constants for surfactant exchange at an air/water interface. *J. Colloid Interface Sci.* **1998**, *205* (2), 213−230.
(7) Jin, F.; Balasubramaniam, R.; Stebe, K. J. Surfactant adsorption to spherical particles: The intrinsic length scale governing the shift from diffusion to kinetic-controlled mass transfer. *J. Adhes.* **2004**, *80* (9), 773−796.
(8) Eastoe, J.; Dalton, J. S. Dynamic surface tension and adsorption mechanisms of surfactants at the air-water interface. *Adv. Colloid Interface Sci.* **2000**, *85* (2−3), 103−144.
(9) Lin, S. Y.; Tsay, R. Y.; Lin, L. W.; Chen, S. I. Adsorption kinetics of C(12)E(8) at the air-water interface: Adsorption onto a clean interface. *Langmuir* **1996**, *12* (26), 6530−6536.
(10) Lin, S. Y.; Mckeigue, K.; Maldarelli, C. Diffusion-Controlled Surfactant Adsorption Studied by Pendant Drop Digitization. *AIChE J.* **1990**, *36* (12), 1785−1795.

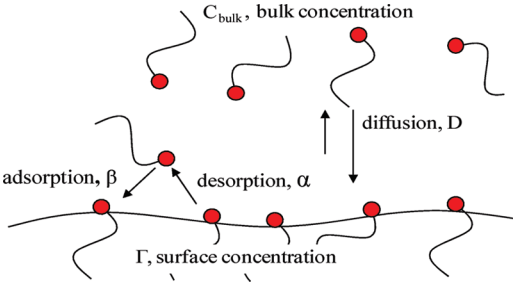



**Figure 1.** Schematic illustration of the three fundamental transport processes (diffusion, adsorption, and desorption) governing surfactant dynamics at an interface.

where $D$ is the diffusion coefficient of the molecule and $h_s$ is given by

$$h_s = b\left(\left(\frac{3h_p}{b}+1\right)^{1/3}-1\right) \tag{2}$$

where $b$ is the bubble radius.[7,11] The planar diffusion time scale is given by

$$\tau_{Dp} = \frac{{h_p}^2}{D} \tag{3}$$

The spherical time scale approaches the planar time scale at large bubble radii and also at large concentrations (i.e., the right-hand side of eq 2 approaches $h_p$ at larger radii). The dependence of diffusive transport on interfacial curvature suggested in eq 1 was only recently confirmed experimentally.[11] Prior to the introduction of this scaling analysis, there has been no systematic study on the dependence of interfacial dynamics on interfacial curvature. We now turn to a discussion of current experimental probes, limitations, and principles on which to design a new apparatus.

In a static drop or bubble suspended in a liquid at the end of a capillary tube, the capillary pressure jump across the interface is proportional to the local curvature of the interface and the surface tension. Hydrostatic pressure variations modify the force balance if the bubble is large enough. The Young−Laplace equation describes this normal stress balance

$$p_i - p_o = (\rho_o - \rho_i)gh + \gamma\nabla\cdot n \tag{4}$$

where $p_i$ is the absolute pressure inside the bubble, $p_o$ is the absolute pressure outside the bubble within the surfactant solution, $\rho_i$ is the density of fluid inside the drop or bubble, $\rho_o$ is the density of the surfactant solution, $\gamma$ is the interfacial tension, $g$ is the acceleration due to gravity, $h$ is the $z$-position along the interface, and $\nabla\cdot n$ is the curvature of the interface. For a spherical bubble, the curvature of the interface is constant, such that $\nabla\cdot n = 2/R$, where $R$ is the radius of the sphere. If we scale eq 4 by normalizing the curvature by the inverse radius of the capillary tube, $1/R_c$, the interface position by the radius of the capillary tube, and the pressure by a characteristic capillary pressure, $\gamma/R_c$, we obtain the dimensionless expression

$$\Delta\overline{p} = B_o\overline{h} + \nabla\cdot\overline{n} \tag{5}$$

where $B_o = \Delta\rho g {R_c}^2/\gamma$ is the Bond number, which describes the relative importance of gravity compared with capillary forces. Equation 5 shows that the interface shape is unique for a given value of the Bond number.

For a drop or bubble pinned at the end of a needle submerged in another fluid at moderate values of the Bond number, the interface shape is no longer spherical and depends only on $\Delta\rho$, $R_c = R_0$ (the radius of curvature at the apex) and $\gamma$. If both $\Delta\rho$ and $R_0$ are known, then the only unknown is surface tension. The pendant drop technique, commonly used to measure surfactant dynamics for both initially clean interfaces and compressed or expanded interfaces, uses this principle to calculate the surface tension from digitally captured drop/bubble shapes.[10,12,13] For typical pendant bubble experiments $R_0 \approx 1.5$ mm and thus $B_0 \approx 0.3$ for an air−water interface. This value of the Bond number is sufficient to perturb the bubble shape away from spherical but not sufficient to detach the bubble from the capillary. The Bond number is usually restricted to a value greater than 0.15 due to limitations of the fitting algorithms.[7] Even though a new algorithm has recently been documented that can accurately measure surface tension from drop shapes an order of magnitude below this limit,[13] the range of bubble sizes (for an air−water interface) is no smaller than 600 $\mu$m using the pendant drop technique.

In this paper, pendant bubble experiments are conducted by imaging an air bubble extended from a J-needle that is submerged in surfactant solution. The images are digitized, and the interface coordinates are extracted. The surface tension is obtained by a nonlinear fit of the Young Laplace equation using a robust nongradient simplex minimization routine and image averaging when appropriate. This numerical method results in measurable shapes for Bond numbers as low as $B_o = 0.01$.[13]

When the Bond number is small, $B_o \ll 0.01$, gravitational forces do not strongly influence the interface shape, and the bubble remains nearly spherical, $R_c = b$. In this case, eq 5 is simplified and the pressure jump across the bubble is given by

$$\Delta p = \frac{2\gamma}{b} \tag{6}$$

where $b$ is the radius of curvature. $B_o \ll 0.01$ can be achieved by either matching the density of the inner and outer fluids, which is common in oil−water systems, or by reducing the radius of the interface. For a microscopic, clean air−water interface, where $b =$ 150 $\mu$m, the value of $B_o = 0.003$ makes the first term in eq 5 negligible. At this value of $B_o$ the interface is spherical, and the pressure jump calculated using eq 6 for a clean air−water interface is $\Delta P = 973$ Pa. Measuring both the pressure jump across the interface and the radius of the spherical interface, the surface tension is calculated directly using eq 6. On the basis of these estimates we have custom-built a "microtensiometer" which allows the measurement of dynamic surface tension from direct measurements of the pressure jump across the interface and the radius of the spherical interface. This apparatus is described in detail in the next section.

The use of the Laplace equation (eq 6) to measure surface tension is not a new concept. The maximum bubble pressure technique measures the maximum pressure required to push a bubble from the tip of a capillary. Since the maximum pressure will occur when the radius of the interface is approximately the radius of the capillary, the radius at this pressure is known and the surface tension is calculated using eq 6.[14,15] The pressure derivative

(11) Alvarez, N. J.; Walker, L. M.; Anna, S. L. Diffusion-limited Adsorption to a Spherical Geometry: The Impact of Curvature and Competitive Timescales. *Phys. Rev. E* **2010**, in press.

(12) Rotenberg, Y.; Boruvka, L.; Neumann, A. W. Determination of Surface Tensions and Contact Angles from the Shape of Axisymmetric Interfaces. *J. Am. Oil Chem. Soc.* **1982**, *59* (4), A297−A297.

(13) Alvarez, N. J.; Walker, L. M.; Anna, S. L. A non-gradient based algorithm for the determination of surface tension from a pendant drop: Application to low Bond number drop shapes. *J. Colloid Interface Sci.* **2009**, *333* (2), 557−562.

(14) Sugden, S. The determination of surface tension from the maximum pressure in bubbles. *J. Chem. Soc.* **1922**, *121*, 858−866.

(15) Sugden, S. The determination of surface tension from the maximum pressure in bubbles Part II. *J. Chem. Soc.* **1924**, *125*, 27−31.

method uses a pressure transducer to measure the pressure jump across the interface and an independent measure of the radius to compute the equilibrium interfacial tension between two pure fluids.[16] This method relies on multiple points taken at different radii of a growing drop to determine the interfacial tension. The growing drop technique uses the instantaneous radius of the drop and pressure jump across the interface to compute the interfacial tension using eq 6.[17] Both the pressure derivative method and the growing drop method use a syringe pump to deliver a constant flow rate of the inner fluid. All these methods are applied to bubbles ranging from 0.3 to 2 mm in radius and have not assessed the dependence of dynamic surface tension on the curvature of the interface.

Other experiments have reported the measurement of dynamic surface tension and equilibrium surface tension using microscopic interfaces. For example, using a constant pressure, Lee et al. constrain a gas−liquid interface inside a tapered glass capillary.[18,19] A second glass capillary delivers a surfactant solution to the interface within the tapered glass capillary. By measuring the radius of the interface and knowing the applied pressure, the surface tension of the interface is determined from the Young−Laplace equation. The authors used this method to measure dynamic surface tension of phospholipids and qualitatively note differences compared with pendant drop measurements.[19] This observation motivated us to fully characterize the impact of radius of curvature on surfactant transport to interfaces. To date, no quantitative study of dynamic surface tension dependence on interface curvature has been performed. The confinement of the interface within the capillary complicates the modeling and characterization of the mass transport of surfactant species since the geometry is finite and depletion effects may become important.[20] Microfluidic methods have also been developed to measure interfacial tension at microscale interfaces, but these methods require the presence of convective bulk flow, which also complicates modeling and characterization.[21] The microtensiometer used in this study considers a simpler geometry, that of a spherical cap immersed in a semi-infinite surfactant solution, and limits measurements to transport by diffusion only.

The principal aims of the present study are to develop a device capable of measuring dynamic surface tension at microscopic interfaces and to quantitatively validate the dependence of dynamic surface tension on the curvature of the interface. The surfactant chosen for this study is $C_{12}E_8$, a well-characterized nonionic surfactant.[9,22,23] The dynamic data will be interpreted in the context of the time scale for diffusion to a spherical interface and a transport model for surfactant adsorption to initially clean and compressed-expanded interfaces.

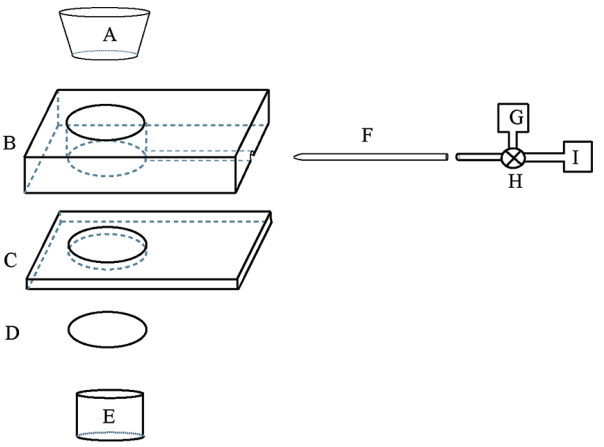



**Figure 2.** Schematic diagram of the microtensiometer apparatus. Parts include (A) microscope condenser, (B) PDMS well and holder, (C) PDMS spacer, (D) no. 1 coverslip, (E) objective and image analysis, (F) glass capillary, (G) pressure transducer, (H) 3-way solenoid valve, and (I) constant pressure head.

## Experimental Section

**Microtensiometer Apparatus.** The key features of the microtensiometer apparatus are shown schematically in Figure 2. The sample cell is fabricated by permanently bonding parts (B), (C), and (D) using radio-frequency ionizing plasma.[24] The capillary (F) is then inserted into the prefabricated hole in (B) and is connected using polyethylene tubing to a three-way solenoid valve (H). The pressure transducer (G) and pressure source (I) are also connected to the three-way valve. The sample cell is designed to fit squarely on a custom horizontal microscope stage. Parts (A) and (E) correspond to the condenser and objective of a Nikon T-300 inverted light microscope.

The sample cell (parts (B)−(D)) is simple and could be fabricated from essentially any material and the dimensions adjusted within some constraints. Cross-linked polydimethylsiloxane (PDMS) is chosen for parts (B) and (C) due to its availability and ease of use. The PDMS is fabricated using a standard kit containing liquid silicone and curing agent in a 10:1 ratio by mass (Dow Sylgard 184 PDMS Kit, purchased from Essex Brownell). The device is formed in a custom mold fabricated from epoxy resin. The overall length and width of part (B) is $2.5 \times 1.5$ in., and the depth is between 0.3 and 0.5 in. The variation in depth allows for different solution volumes. The diameter of the well is 1.25 cm, and the groove on the bottom is 1 mm in diameter. The dimensions of the well are chosen to ensure that adsorption of surfactant to solid−liquid interfaces (PDMS and glass) does not deplete enough surfactant to significantly reduce the bulk concentration. Part (C) is a copy of part (B), except that the depth is between 1 and 3 mm and there is no groove. The depth of part (C) is adjustable to ensure a semi-infinite solution volume surrounding the interface. Experiments are performed at different depths of the capillary until the dynamic response is observed to be independent of distance from the bottom glass coverslip (D). Part (D) is a 1 in. no. 1 circular glass coverslip (Warner Instruments, Hamden, CT). The same type of coverslip is also used to cover the top of the cell during experiments to minimize the effects of evaporation and convection.

The capillary (F) is either purchased or fabricated using custom settings on a P-80 capillary puller (Sutter Instrument Co., Novato,

(16) Passerone, A.; Liggieri, L.; Rando, N.; Ravera, F.; Ricci, E. A New Experimental-Method for the Measurement of the Interfacial-Tension between Immiscible Fluids at Zero Bond Number. *J. Colloid Interface Sci.* **1991**, *146* (1), 152−162.

(17) Macleod, C. A.; Radke, C. J. A Growing Drop Technique for Measuring Dynamic Interfacial-Tension. *J. Colloid Interface Sci.* **1993**, *160* (2), 435−448.

(18) Lee, S.; Kim, D. H.; Needham, D. Equilibrium and dynamic interfacial tension measurements at microscopic interfaces using a micropipet technique. 1. A new method for determination of interfacial tension. *Langmuir* **2001**, *17* (18), 5537−5543.

(19) Lee, S.; Kim, D. H.; Needham, D. Equilibrium and dynamic interfacial tension measurements at microscopic interfaces using a micropipet technique. 2. Dynamics of phospholipid monolayer formation and equilibrium tensions at water-air interface. *Langmuir* **2001**, *17* (18), 5544−5550.

(20) Filippov, L. K.; Filippova, N. L. Dynamic surface tension and adsorption kinetics in finite systems. *J. Colloid Interface Sci.* **1997**, *187* (2), 352−362.

(21) Martin, J. D.; Hudson, S. D. Mass transfer and interfacial properties in two-phase microchannel flows. *New J. Phys.* **2009**, *11*, 115005.

(22) Tsay, R. Y.; Lin, S. Y.; Lin, L. W.; Chen, S. I. Adsorption kinetics of C12E8 at the air-water interface: Desorption from a compressed interface. *Langmuir* **1997**, *13* (12), 3191−3197.

(23) Liggieri, L.; Ferrari, M.; Massa, A.; Ravera, F. Molecular reorientation in the adsorption of some CiEj at the water-air interface. *Colloids Surf., A* **1999**, *156* (1−3), 455−463.

(24) Duffy, D. C.; McDonald, J. C.; Schueller, O. J. A.; Whitesides, G. M. Rapid prototyping of microfluidic systems in poly(dimethylsiloxane). *Anal. Chem.* **1998**, *70* (23), 4974−4984.

CA) or a PMP-100 capillary puller (Micro Data Instrument Inc., South Plainfield, NJ). Capillary diameters ranging from 12 to 300 $\mu$m are used in the experiments reported here. Capillaries used for pulling are thin-walled capillaries purchased from WPI Inc. (Sarasota, FL), with dimensions of i.d. = 0.75 mm, o.d. = 1 mm, and $L = 4$ in. To avoid wetting issues, the capillaries are cleaned by first submerging them in sulfuric acid and then rinsing with acetone. This process is repeated until an air bubble can be pinned at the edge of the capillary tip. The capillaries are also tested for axisymmetry by measuring the radius of a bubble suspended at the tip at different angles of rotation about the longitudinal axis of the capillary. Capillaries with excessive asymmetry, indicated by a difference in surface tension greater than 1 mN/m as a function of rotation, are discarded.

The bubble or drop is suspended at the end of the capillary using a constant pressure head. The pressure head is generated using a column of water attached by polyethylene tubing to the three-way solenoid valve. The pressure head is measured using two strain-gauge pressure transducers: an Omega Model PX409-001GV for pressures up to 1 psi (6895 Pa) and a Cole Parmer Model EW-68075-10 for pressures up to 5 psi (34 474 Pa). The pressure transducers are calibrated using a water column with an accuracy of 0.063 in. of $H_2O$ (15 Pa). The pressure readings are recorded using a National Instruments Fieldpoint Module AI-100.

The interface is imaged using a Diagnostic Instruments Spot RT Monochrome digital camera connected to the side port of the microscope. Nikon ELWD Plan Fluoro microscope objectives are used with magnifications of 10×, 20×, and 40×. The camera and optical setup are calibrated for all microscope objectives using a calibration slide with 10 $\mu$m graduations. The 10×, 20×, and 40× objectives are measured to have calibration factors of 5.0 ± 0.09, 9.8 ± 0.2, and 194 ± 0.3 pixels/$\mu$m, respectively. The image is captured using a custom Labview subroutine and subsequently analyzed *in situ* by fitting a circle to the extracted interface image. The National Instruments Vision toolbox is used for interface detection and fitting.

Pressure and radius are measured simultaneously and surface tension is calculated using

$$\gamma(t) = \frac{1}{2}\Delta P(t) R(t) \tag{7}$$

where $\Delta P(t)$ is the pressure jump across the interface. The pressure jump is computed by subtracting the hydrostatic pressure head of the solution acting at the capillary tip from the measured constant pressure head. The maximum frame rate and corresponding data acquisition rate is 6.5 frames/s. Although we are currently restricted to 6.5 frames/s, by using a high-speed camera the data acquisition rate could be increased up to 100 000 frames/s. Bubbles can be formed on millisecond time scales in either the microtensiometer or a pendant drop apparatus; however, at the length scales of the pendant drop apparatus, this would lead to significant convection, complicating dynamic surface tension measurements.[25] The length scales in the microtensiometer apparatus should alleviate this problem. Therefore, the microtensiometer has the ability to measure subsecond dynamic surface tension, allowing for faster dynamics to be characterized.

The microtensiometer apparatus is validated using dynamic surface tension measurements of well-characterized pure fluid−fluid systems. For example, the measured surface tension at the interface of an air bubble in clean water is 73.2 ± 0.4 mN/m, and that of an air bubble in clean ethanol is 22.7 ± 0.3 mN/m. The results given represent average values of several measurements taken as a function of time using capillaries of different radii (40−150 $\mu$m) at 20 °C. The reported values of the interfacial tension for both liquids are 72.9 and 22.4 mN/m.[26] The uncertainty in the calculated surface tension is less than $\Delta\gamma < \pm 0.5$ mN/m for the two measured fluid−liquid pairs and represents the expected error for dynamic measurements involving surfactants.

Analysis of the propagation of errors shows that the largest random errors occur at the smallest (17 $\mu$m) and largest (150 $\mu$m) capillary radii. The uncertainty for the smallest capillaries is dominated by random errors associated with fitting the interface radius, while the random errors in the pressure measurement are most important for the largest capillaries. The estimated uncertainty for the smallest radius is ±0.2 mN/m. The estimated uncertainty for the largest radius is ±0.5 mN/m. Note that the measured experimental random error is similar to the error contributions estimated from propagation of error analysis, suggesting that there are no major contributions to error other than pressure and radius.

Two types of experiments are conducted using the microtensiometer apparatus. The first experiment measures the evolution of surface tension for an initially clean interface. We interpret the change in surface tension as the result of surfactant populating the interface from the bulk solution. For this experiment air is passed through the needle at a relatively large flow rate to purge any adsorbed surfactant from the interface. The flow is then stopped, and the interface is held at constant pressure while the interface radius is measured as a function of time. The second experiment measures the evolution of surface tension of an interface that is instantaneously expanded or compressed after reaching equilibrium. We interpret the change in surface tension in this case as a re-equilibration of the surface concentration with the bulk surfactant concentration. A solenoid valve is used to subject the interface to a different pressure head, which leads the interface to expand if the pressure is increased or to compress if the pressure is decreased (cf. eq 6). The surface tension is monitored from the instant that the step change in surface area is applied until equilibrium is re-established.

In this apparatus, because the bubble is maintained using a constant pressure head, the change in surface tension is primarily accounted for by a change in radius; the radius of the interface continuously changes in a dynamic experiment. The dynamics depend on radius and therefore to properly model the behavior observed in the experiments the changing radius must be taken into account. We account for this effect in a numerical model that we will describe in a later section. The flow generated by the changing radius is neglected. The maximum velocity corresponding to the changing radius is estimated from experimental data and is less than $V_{max} < 1 \times 10^{-6}$ m/s. The maximum velocity occurs at the apex of the spherical cap. This calculation is performed using the height equation for a spherical cap, and the maximum velocity is estimated from the change in height of the cap with time. The minimum velocity occurs at the fixed contact line where the bubble is pinned at the edge of the capillary tube. The experimental velocity reported above is estimated by tracking the height of the observed spherical interface as a function of time and then computing the derivative of the time-dependent height. The estimated velocities are small enough that we conclude convection effects due to changing radius are negligible.

If the bubble radius continuously changes during an experiment, then so does the interfacial area. Experimental data are used to estimate whether the changing surface area significantly influences the evolution of the surface concentration. Figure 3 shows the ratio of the surface area at a given time to the surface area for the previous time step as a function of time for the capillary tip with the largest radius (150 $\mu$m) and a relatively large surfactant concentration (6.0 $\mu$M). The area is calculated using the equation for the surface area of a spherical cap pinned at the end of a capillary of fixed radius. The results shown in Figure 3 demonstrate that the change in area is less than 0.1% for any given time step. Therefore, even though the change in radius is taken into account because diffusion to the interface depends on the magnitude of the interface radius (cf. eq 1), we conclude that there is no appreciable effect on the surface concentration evolution, and the change in surface area is neglected in our theoretical analysis.

(25) Moorkanikkara, S. N.; Blankschtein, D. Possible Existence of Convective Currents in Surfactant Bulk Solution in Experimental Pendant-Bubble Dynamic Surface Tension Measurements. *Langmuir* **2009**, *25* (3), 1434−1444.

(26) Adamson, A. W.; Gast, A. P. *PhysicalChemistry of Surfaces*, 6th ed.; Wiley: New York, 1997; pp XXI, 784 s.

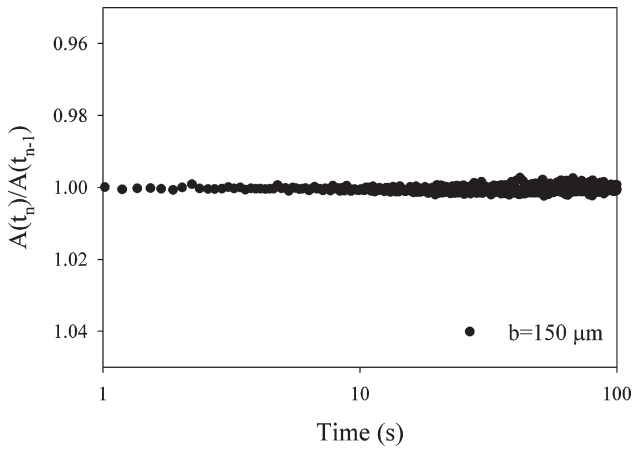



**Figure 3.** Ratio of change in area as a function of time for a dynamic surface tension experiment conducted on a 150 $\mu$m radius bubble and $C_{bulk}$ = 6.0 $\mu$M. The ratio is essentially unity for all times.

A well-characterized nonionic surfactant, $C_{12}E_8$, is chosen for this study. The primary goal of this paper is to determine the dependence of dynamic surface tension on interfacial radius in the absence of convection and to make a comparison with documented measurements on large radius bubbles.[9,22,23,27−30]

**Materials.** Solvents (ACS grade) for cleaning and preparation were purchased and used as received. The nonionic surfactant polyoxyethylene glycol alkyl ether $C_{12}E_8$ ($C_{12}H_{25}O(CH_2CH_2O)_8H$) was purchased from Nikko Chemicals (99% purity) and used as received. Surfactant solutions were prepared using deionized water purified using a Barnstead UV Ultrapure II purification system (resistivity of 18.2 MΩ·cm). $C_{12}E_8$ solutions are prepared by first melting the pure surfactant at 40 °C and then weighing a known amount of pure surfactant. We selected three values of the bulk concentration below the critical micelle concentration (cmc = 100.0 $\mu$M at 20 °C): $C_{bulk}$ = 1.0, 2.5, and 6.0 $\mu$M.

## Numerical Analysis

To compare experiments with a transport model, a numerical scheme is developed to integrate the governing equations. Surfactant adsorption/desorption to a spherical interface is modeled using Fick's law of diffusion and a kinetic rate equation. Diffusion to a spherical interface is given by

$$\frac{\partial C}{\partial t} = \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C}{\partial r}\right) \tag{8}$$

The kinetic rate equation that is most often used to describe nonionic surfactants, such as the $C_{12}E_8$ surfactant studied here, is the generalized Frumkin equation, given by

$$\frac{\partial \Gamma}{\partial t} = \beta C_s \Gamma_\infty \left(1 - \frac{\Gamma}{\Gamma_\infty}\right) - \alpha\Gamma \exp\left(\kappa\left(\frac{\Gamma}{\Gamma_\infty}\right)^n\right) \tag{9}$$

where $\beta$ is the adsorption constant, $\alpha$ is the desorption constant, $\Gamma_\infty$ is the maximum packing concentration of surfactant at the interface, $\kappa$ is said to account for molecular interactions, and $n$ is a fitting parameter accounting for the nonlinearity of the interactions. The relevant boundary conditions are a flux balance at the interface between the surface concentration and the bulk fluid concentration immediately adjacent to the interface. This flux is given by

$$\frac{\partial \Gamma}{\partial t} = D\left.\frac{\partial C}{\partial r}\right|_{r=b} \tag{10}$$

The second boundary condition is such that far from the interface $\lim_{r\to\infty} C \to C_{bulk}$. If eqs 8 and 9 are solved subject to these boundary conditions, all modes of surfactant transport to the interface are described. Two special cases of this solution are diffusion-limited and kinetic-limited dynamics. The diffusion-limited case can be described by simultaneously solving eqs 8 and 9, except that the right-hand side of eq 9 is set equal to zero. In other words, at all time steps the subsurface and the interface concentrations are in equilibrium. The kinetic-limited case can be solved by integrating eq 9 only. To describe surfactant transport onto a freshly formed interface, we use the following initial conditions:

$$\Gamma(t = 0) = 0 \quad \text{and} \quad C(t = 0, r) = C_{bulk} \tag{11}$$

To describe surfactant transport onto an expanded or compressed interface, we use the following initial conditions:

$$\Gamma(t = 0) = \Gamma_{eq}(C_{bulk}) \quad \text{and} \quad C(t = 0, r) = C_{bulk} \tag{12}$$

To solve this problem, we employ a collocation spectral method in which space is discretized using Chebyshev gridpoints and time is discretized using an implicit Euler scheme.[31] To ensure that convergence is met, we changed the number of grid points and the time step until the solution showed a less than $10^{-6}$ deviation in $\gamma(t)$ with a change in these parameters. In addition, we validated the numerical procedure, using a similar method described in the literature,[32] by integrating the spherical Ward and Tordai equation[10]

$$\Gamma(t) = \Gamma_i + \frac{D}{b}\left[C_{bulk}t - \int_0^t C_s(\tau)\,d\tau\right] + \left(\frac{4D}{\pi}\right)^{1/2}\left[C_{bulk}\sqrt{t} - \int_0^{\sqrt{t}} C_s(t-\tau)\,d\sqrt{\tau}\right] \tag{13}$$

using the trapezoidal rule, where $\Gamma_i$ is the initial surfactant coverage on the interface.

As explained in the Experimental Section, the bubble undergoes a change in radius with time during an experiment since pressure is kept relatively constant during a dynamic surface tension measurement. The dynamics are highly dependent on curvature, and this changing radius should be accounted for to correctly model the experimental data. The experimental pressure data are introduced in the numerical solution in order to accomplish this. First, a spline curve is fit to the measured transient pressure data. Then the numerical calculation is initiated using the initial radius of the experiment, and the surface tension is calculated at the subsequent time step, $t_n$. The resulting surface tension coupled with the corresponding pressure interpolated at the given time step, $t_n$, is used to calculate the radius, $b$, at the current time step. This radius is then used in eq 13 to calculate the

(27) Miller, R.; Fainerman, V. B.; Mohwald, H. Adsorption behavior of oxyethylated surfactants at the air/water interface. *J. Colloid Interface Sci.* **2002**, *247* (1), 193−199.
(28) Lin, S. Y.; Lee, Y. C.; Yang, M. W.; Liu, H. S. Surface equation of state of nonionic CmEn surfactants. *Langmuir* **2003**, *19* (8), 3164−3171.
(29) Lee, Y. C.; Lin, S. Y.; Liu, H. S. Role of equation of state on studying surfactant adsorption kinetics. *Langmuir* **2001**, *17* (20), 6196−6202.
(30) Lee, Y. C.; Stebe, K. J.; Liu, H. S.; Lin, S. Y. Adsorption and desorption kinetics of CmE8 on impulsively expanded or compressed air-water interfaces. *Colloids Surf., A* **2003**, *220* (1−3), 139−150.
(31) Trefethen, L. N. *SpectralMethods in MATLAB*; Society for Industrial and Applied Mathematics: Philadelphia, 2000; pp XVI, 165 s.
(32) Miller, R.; Kretzschmar, G. Numerical-Solution for Mixed Model of Diffusion Controlled Adsorption. *Colloid Polym. Sci.* **1980**, *258* (1), 85−87.

surface tension at the next time step, $t_{n+1}$. This process is repeated until the surface tension reached equilibrium or the loop is terminated at a specified time. The numerical analysis presented here will be compared to our experimental results.

## Results

Relaxation of surface tension with time due to adsorption of $C_{12}E_8$ is measured using the microtensiometer for an initially clean air−water interface, and the results are shown in Figure 4. Figure 4a shows the surface tension evolution as a function of time for a fixed $C_{bulk} = 1.0\,\mu$M (1.0% cmc) and different capillary radii: $b = 17, 30, 45, 70, 130\,\mu$m, and a pendant bubble. Pendant bubble measurements are taken using a separate device.[13] Clean bubbles are formed by pushing a steady stream of air out of the capillary so that all dynamic surface tension curves start at the clean interfacial tension value for air and water, $\gamma_0 = 73.2 \pm 0.25$ mN/m, and decrease to an equilibrium surface tension value of $\gamma_{eq} = 62.4 \pm 0.3$ mN/m. The radius of the pendant bubble is ~1200 $\mu$m. Both devices yield the same equilibrium surface tension values, to within the expected uncertainty observed in our experiments. The time scales for reaching equilibrium change significantly with bubble radius. As bubble radius decreases, the dynamics become faster and the time to reach equilibrium is reduced. For example, it takes ~1000 s for the surface tension to reach the halfway point (68 mN/m) in the pendant bubble apparatus and only 50 s to reach the same point for the 17 $\mu$m radius microtensiometer experiments.

Figure 4b shows the surface tension as a function of time for $C_{bulk} = 2.5\,\mu$M (2.5% cmc) measured at capillary radii of $b = 17, 40, 70, 150\,\mu$m, and a pendant bubble. As before, the curves begin at the clean interfacial tension value and decrease to an equilibrium value of $\gamma_{eq} = 57.8 \pm 0.3$ mN/m. Comparing Figure 4a to Figure 4b, it is observed that the equilibrium surface tension decreases as the bulk concentration increases, as expected. In addition, the dynamics become faster as concentration increases. Figure 4c shows the surface tension as a function of time for $C_{bulk} = 6.0\,\mu$M (6.0% cmc) measured at capillary radii of $b = 17, 40, 70, 150\,\mu$m, and a pendant bubble. The equilibrium surface tension value for this concentration is $\gamma_{eq} = 53.8 \pm 0.3$ mN/m. It is evident Figure 4a−c that the surface tension evolves significantly faster for bubbles tens of micrometers in diameter compared with millimeter-scale bubbles. The equilibration time is reduced by an order of magnitude from the largest radius, 1200 $\mu$m, to the smallest radius, 17 $\mu$m.

Figure 5 shows equilibrium surface tension as a function of bulk surfactant concentration for $C_{12}E_8$ obtained using the microtensiometer at 20 ± 0.5 °C. Equilibrium surface tension values are extracted from the long time asymptotes of dynamic surface tension measurements as seen in Figure 4. The generalized Frumkin (GF) parameters are fit using a reformulation of the nonlinear least-squares analysis presented previously for the purpose of fitting isotherms.[13] The best-fit parameters are given in Table 1 along with parameter values obtained from the literature[9,27] for the same surfactant at 25 °C. Since care is taken to ensure that there are no depletion effects in the sample cell, the difference between literature values and values obtained here is likely due to the 5 °C temperature difference between studies. Another isotherm argued to be appropriate for $C_iE_j$ surfactants, known as the reorientation model,[33] is also fit to the isotherm data shown in Figure 5. This isotherm will be discussed further later in the text. The best fit parameters for this isotherm are also given in Table 1.

(33) Fainerman, V. B.; Makievski, A. V.; Miller, R. Surfactant adsorption isotherms considering molecular reorientation or aggregation at liquid/fluid interfaces. *Rev. Chem. Eng.* **1998**, *14* (6), 373−407.

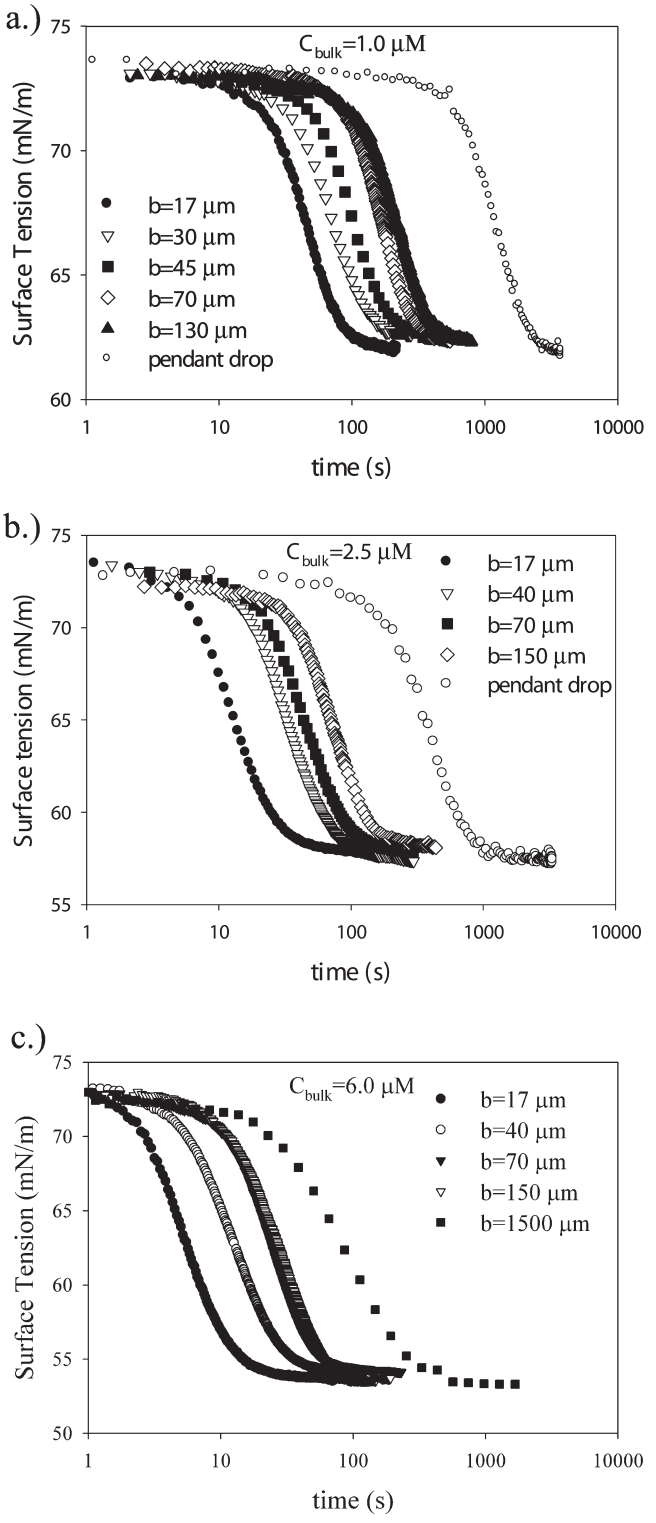



**Figure 4.** Surface tension as a function of time for different radii of curvature at fixed surfactant concentrations. (a), (b), and (c) correspond to experiments at bulk concentrations of 1.0, 0.25, and 6.0 $\mu$M, respectively. PB refers to pendant bubble experiments.

By allowing the interface to achieve equilibrium and then rapidly increasing or decreasing the interfacial area, the relaxation of surface tension due to a perturbation about the equilibrium surface concentration is probed. Figure 6 shows the relaxation of the normalized surface tension

$$\theta = (\gamma(t) - \gamma_{eq})/(\gamma_0 - \gamma_{eq}) \quad (14)$$

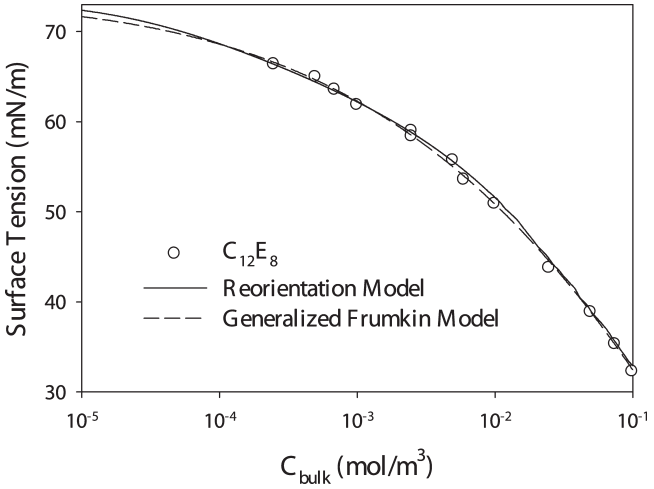



**Figure 5.** Equilibrium surface tension as a function of concentration (i.e., isotherm data) for $C_{12}E_8$ at 20 °C. The two lines correspond to the best fit parameters for the generalized Frumkin isotherm and the reorientation isotherm.

**Table 1. A Comparison between Best Fit Parameters from Equilibrium Data for the Generalized Frumkin Isotherm (Top) and Reorientation Isotherm (Bottom) Obtained at 20°C Using a New Microtensiometer Apparatus with Those from the Literature Obtained at 25 °C**

| isotherm parameters | $\Gamma_\infty (10^{-6})$ [m$^2$/mol] | $a\,(10^{-6})$ [mol/m$^3$] | $\kappa$ | $n$ |
|---|---|---|---|---|
| 20 °C | 6.5 | 3.87 | 11.7 | 0.381 |
| 25 °C[9,28] | 5.3 | 2.33 | 13.2 | 0.503 |
| isotherm parameters | $\omega_1(10^5)$ [mol/m$^2$] | $\omega_2(10^5)$ [mol/m$^2$] | $b_o(10^3)$ [m$^3$/mol] | $\alpha_o$ |
| 20 °C | 9.45 | 2.74 | 0.852 | 2.67 |
| 25 °C[27] | 10.0 | 3.42 | 2.34 | 2.75 |

The expansion experiments correspond to positive displacements from equilibrium, while the compression experiments correspond to negative displacements. Expansion of the interface results in a reduced surface concentration since the area of the interface increases while the number of adsorbed molecules stays the same. The opposite is true for compression experiments, where the area decreases and the number of molecules stays the same. Three bulk concentrations are shown in the figure, corresponding to $C_{bulk}$ = 0.6, 2.5, and 6.0 $\mu$M from left to right for both expansion and compression experiments. A wide range of experiments are conducted at various bubble radii and surfactant concentrations. The results shown here are experiments conducted at a fixed expansion and compression ratio. The radii used for the expansion experiments are 22, 38, and 38 $\mu$m from left to right and for compression 24, 45, and 24 $\mu$m from left to right. For the fixed radius expansion and compression experiments performed at 38 and 24 $\mu$m, respectively, as concentration decreases the relaxation of surface tension is slower for both types of experiments, so that the curves shift to the right as concentration decreases. For the expansion experiments the change in surface area is $A_0/A$ = 0.833, 0.832, and 0.827, respectively. The compression area ratios are $A_0/A$ = 1.27, 1.25, and 1.23, respectively.

This new apparatus is successfully used to measure the dependence of surfactant dynamics on bubble radius for initially clean interface dynamics as well as measure dynamic re-equilibration after a sudden expansion or compression. The initially clean interface measurements show a strong dependence of surfactant dynamics on radius of curvature. Namely, as radius decreases, the equilibration time decreases. There is good agreement between

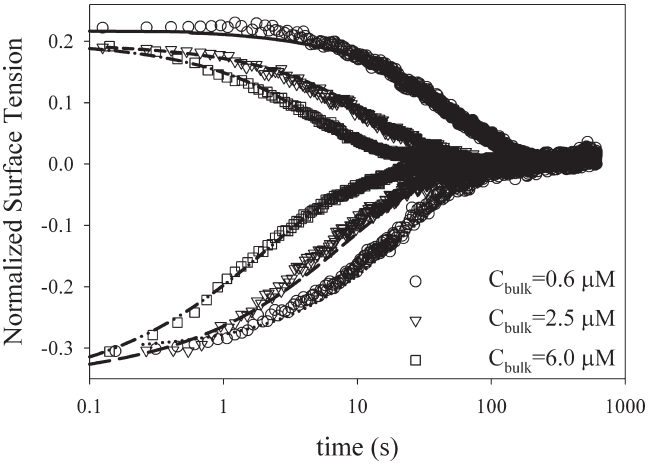



**Figure 6.** Normalized compression/expansion experiments for different concentrations and bubble radii. Compression experiments correspond to a relaxation from a negative normalized surface tension, and expansion experiments correspond to positive values. The bulk surfactant concentration corresponding to each curve from left to right are $C_{bulk}$ = 0.6, 2.5, and 6.0 $\mu$M. The expansion experiments correspond to a change in surface area of $A_0/A$ = 0.833, 0.832, and 0.827 from left to right. The compression area ratios are $A_0/A$ = 1.27, 1.25, and 1.23 from left to right. The lines correspond to model predictions. The molecular diffusion coefficient $D = 3.8 \times 10^{-10}$ m$^2$/s and the generalized Frumkin isotherm are used for these predictions.

the equilibrium surface tension values obtained with the microtensiometer and the pendant bubble apparatus. However, the microtensiometer uses significantly less volume than the pendant bubble and equilibrium is reached more rapidly.

Characteristic time scales for each dynamic surface tension experiment are shown in Figure 7. For each experiment, the time corresponding to a given surface coverage $\phi = \Gamma/\Gamma_{eq}$ is extracted from the data shown in Figure 4a−c. The open symbols correspond to a surface coverage of $\phi$ = 0.6 and the closed symbols to $\phi$ = 0.1. The experiment time is normalized by the characteristic time for diffusion to a planar interface (eq 3) and is plotted as a function of the capillary radius normalized by the intrinsic depletion depth $h_p$. To scale the data, an isotherm is required to calculate the depletion depth and the planar diffusion time scale. The generalized Frumkin isotherm with the fitted parameters corresponding to our measured data as given in Table 1 is used to perform the scaling analysis. In Figure 7, the solid line represents the normalized characteristic time scale for diffusion to a spherical interface (eq 1), and the dashed line corresponds to that of a planar interface. The data and analysis presented in Figure 7 are previously presented,[11] but here the data are presented to emphasize the dependence on bubble radius and concentration for each data set. Figure 7 shows that as the concentration increases, the experiment time scale approaches the planar time scale, and as the radius decreases, the experiment time scale is smaller than the planar time scale, i.e., the dynamics are faster for smaller radii.

The scaled data show very good agreement with the time scale for diffusion to a spherical interface. If kinetic processes of adsorption and desorption are important, then there will be a deviation from the diffusion time scaling.[11] The agreement between the experimental data and the diffusion time scale in Figure 7 strongly suggests that $C_{12}E_8$ undergoes diffusion-limited transport for all experimental conditions considered here. This result is contrary to current literature and shows that kinetic exchange at the interface does not need to be invoked to capture the dynamics of this surfactant. The advantage of knowing that the surfactant dynamics follow diffusion-limited

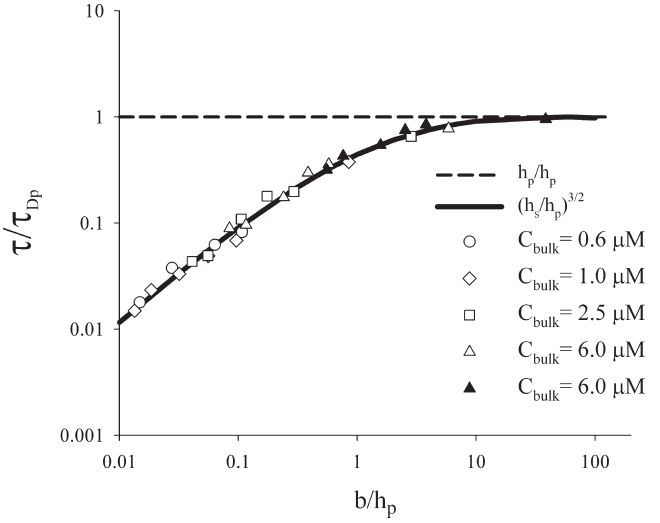



**Figure 7.** Scaled experimental time as a function of $b/h_p$ for different concentrations and bubble radii. The symbols correspond to experimental time scales extracted from Figure 4.

transport is that a lower limit on $\beta$, the adsorption coefficient, can be determined from an asymptotic analysis. This analysis is further explained in Alvarez et al.[11] For example, for $C_{12}E_8$, we previously found that $\beta_{limit} = 5.6$ m$^3$/(mol s).[11]

The ratio of the diffusion time scale to the kinetic time scale (assuming Langmuirian kinetics for simplicity) is set equal to unity, and the radius, $R_{crit}$, where the time scales are equal, is determined by solving

$$\left(R_{crit}\left[\left(\frac{3h_p}{R_{crit}}+1\right)^{1/3}-1\right]\right)^3 h_p(\beta C_{bulk}+\alpha) = 1 \qquad (15)$$

for $R_{crit}$. Scaling by $R_{DK} = D/(\beta\Gamma_\infty)$, an intrinsic length scale introduced by Jin et al.,[7] the new critical radius is given by

$$\frac{R_{crit}}{R_{DK}} = \frac{2}{\sqrt{12-3q^2}-3q} \qquad (16)$$

where $q = (R_{DK}/h_p)^{1/3}$ again assuming Langmuirian kinetics. Figure 8 shows an operating diagram for dynamic surface tension measurements in terms of bubble radius and the bulk concentration of surfactant. The critical radius denoting the transition from diffusion-limited to kinetic-limited transport is plotted using eq 16 for different values of the kinetic adsorption constant. Note that the values chosen for $\beta$ span the bounds put forth in previous studies.[34] The solid lines correspond to the radius at which the two time scales are equal for a given surfactant concentration. Values far to the left of the solid line correspond to diffusion-limited dynamics, and values far to the right correspond to kinetic-limited dynamics. Figure 8 shows the expected trend that as bubble radius decreases, the dynamics become more kinetic-limited. In addition, the figure captures the experimentally and theoretically investigated dependence of the controlling mechanism on surfactant concentration. As concentration increases, the system moves toward the kinetic-limited regime.

The two shaded regions in Figure 8 correspond to the physical operating space for the pendant drop apparatus and the microtensiometer device presented in this paper. From this figure, we

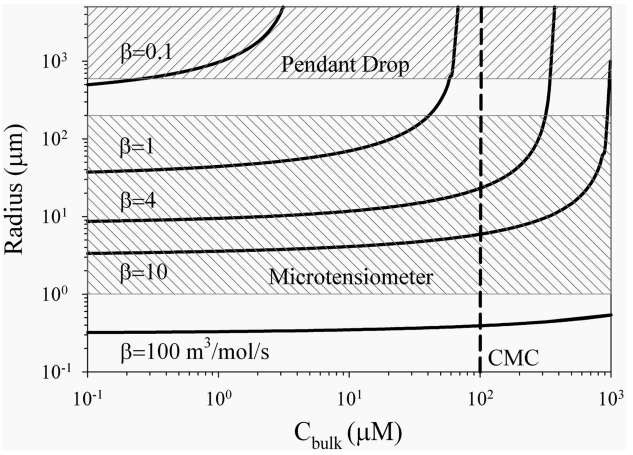



**Figure 8.** Operating diagram showing bubble radius and bulk surfactant concentration. The solid lines correspond to the points when the kinetic and diffusion time scales are equal for a given adsorption constant, $\beta$. Far to the left of the solid line defines diffusion-limited dynamics and far to the right of the line defines kinetic-limited dynamics. The two shaded regions correspond to the experimental operating regimes for the pendant drop/bubble and the microtensiometer apparatus.

conclude that the pendant drop apparatus cannot easily be used to observe kinetic-limited dynamics except for cases when the adsorption coefficient is very small (i.e., $\beta \leq 0.1$ m$^3$/(mol s)). Experiments could be conducted at high concentrations, but the dynamics become too fast to observe and the critical micelle concentration acts as an upper limit for concentration, beyond which dynamics are complicated by the existence of micelles in the bulk solution. Therefore, it is advantageous to work with microscale interfaces and moderately dilute concentrations to observe kinetic-limited transport. Although the results from this study reveal that the dynamics for $C_{12}E_8$ do not diverge from diffusion-limited dynamics, Figure 8 yields information concerning the lower limit on the adsorption coefficient $\beta$ that can be tested in a given apparatus (i.e., for $C_{12}E_8$ $\beta \leq 4$ m$^3$/(mol s) for the microtensiometer and $\beta \leq 0.1$ m$^3$/(mol s) for the pendant drop apparatus).

The scaling analysis presented previously[11] is used to show that $C_{12}E_8$ follows diffusion-limited dynamics for all radii and concentrations tested using the pendant drop apparatus and the microtensiometer. This is made clear by the excellent agreement between the scaled experimental data and the spherical diffusion-limited time scaling. Therefore, to correctly describe the evolution of the surface tension dynamics for $C_{12}E_8$, a diffusion model is required and kinetic exchange at the interface does not need to be considered. In the next section, we compare experiments with a diffusion-limited transport model.

## Prediction of Dynamic Surface Tension

To determine whether the diffusion-limited generalized Frumkin (GF) model correctly predicts the dynamic surface tension response of $C_{12}E_8$ at different concentrations and radii, we used the numerical procedure discussed earlier. Here, we have chosen to use a fixed value of the diffusion coefficient that corresponds to the molecular diffusion coefficient. NMR studies performed on $C_8E_4$ determined that the self-diffusion coefficient for this molecule is $(4.83 \pm 0.12) \times 10^{-10}$ m$^2$/s in $D_2O$.[35] Using the relationship that the diffusion coefficient scales with $M^{-1/2}$, where $M$ is the

(34) Moorkanikkara, S. N.; Blankschtein, D. New methodology to determine the rate-limiting adsorption kinetics mechanism from experimental dynamic surface tension data. *J. Colloid Interface Sci.* **2006**, *302* (1), 1−19.

(35) Faucompre, B.; Lindman, B. Self-Association of Zwitterionic and Non-ionic Surfactants - Nmr Self-Diffusion Studies. *J. Phys. Chem.* **1987**, *91* (2), 383−389.

molecular weight of the surfactant molecule and accounting for viscosity and temperature differences between $D_2O$ and water, the diffusion coefficient for $C_{12}E_8$ is $(3.93 \pm 0.12) \times 10^{-10}$ m$^2$/s. This value is consistent with a previous study in which the authors state that the diffusion coefficient for $C_{12}E_8$ monomer is $3.5 \times 10^{-10}$ m$^2$/s in $D_2O$ at 25 °C.[36] Using the Stokes−Einstein relationship and accounting for both the temperature and the viscosity, the diffusion coefficient of $C_{12}E_8$ is $3.8 \times 10^{-10}$ m$^2$/s in $H_2O$ at 20 °C. The value obtained from the short time analysis of dynamic surface tension yields a diffusion coefficient of $(3.67 \pm 1.6) \times 10^{-10}$ m$^2$/s.[34] The diffusion coefficient used in the present study is $D = 3.8 \times 10^{-10}$ m$^2$/s at 20 °C, based on an average of the estimated values from all three studies.

The first comparison is made between measurements of dynamic surface tension at initially clean interfaces and predictions using the molecular diffusion coefficient and the GF isotherm numerical model. Figure 9 compares experiments and predictions for three different radii for each surfactant concentration. Figure 9a shows predictions for three bubble radii, $b$ = 17, 130 $\mu$m, and a pendant drop at fixed $C_{bulk}$ = 1.0 $\mu$M. At this concentration, the GF model predicts the dynamics for $b$ = 17 $\mu$m well but deviates from the experimental curves for larger radii. Figure 9b and Figure 9c compare model predictions with experiments for $b$ = 17 and 150 $\mu$m and a pendant drop at fixed $C_{bulk}$ = 2.5 $\mu$M and $C_{bulk}$ = 6.0 $\mu$M, respectively. In both cases, the GF model predicts much slower dynamics for $b$ = 17 $\mu$m than observed. For the larger radii, $b$ = 150 $\mu$m and a pendant drop, both models predict similar equilibration time scales to those observed in the experiments but differ in shape. Note that in all predictions the equilibration time, i.e., the time it takes for the dynamics to reach $\gamma_{eq}$, is close to the measured equilibrium time even though the shapes of the predicted and measured curves may differ.

One possible source of the discrepancy in these comparisons is that the selected isotherm is not adequate to capture the dynamics. To examine this possibility, we tested another isotherm given in the literature, known as the "reorientation isotherm", which has been suggested to be appropriate for $C_{12}E_8$.[27] A numerical analysis is conducted using the reorientation model[33] and eq 13. As shown by the dashed lines in Figure 9a−c, use of the reorientation model did not lead to improved agreement between the predictions and data. We also note the agreement cannot be improved by introducing adsorption/desorption kinetic barriers in the model. For a fixed molecular diffusion coefficient, introducing kinetic barriers would change the shape of the curve, shifting the calculated dynamics further to the right of the experimental curves and increasing the predicted equilibration times. One possible reason that a single isotherm cannot capture the observed dynamics is that there is a phase transition from a gaslike to liquidlike state at low surface concentrations.[6,37]

A situation like this could be avoided by using compression−expansion experiments to perturb the interface about the equilibrium concentration. In a compression experiment, the adsorbed species are further pushed into a liquid regime, and therefore no transition of states is expected. In an expansion experiment, as long as the surface concentration is not pushed too far from equilibrium, then the surfactant at the interface should also remain in a liquid state. Thus, compression−expansion can be modeled by any single phase isotherm that accurately fits the measured equilibrium surface tension as a function of bulk concentration. An additional advantage of this type of experiment is that changes in radius are small since the surface tension is only changing by a few mN/m for a given expansion or compression ratio.

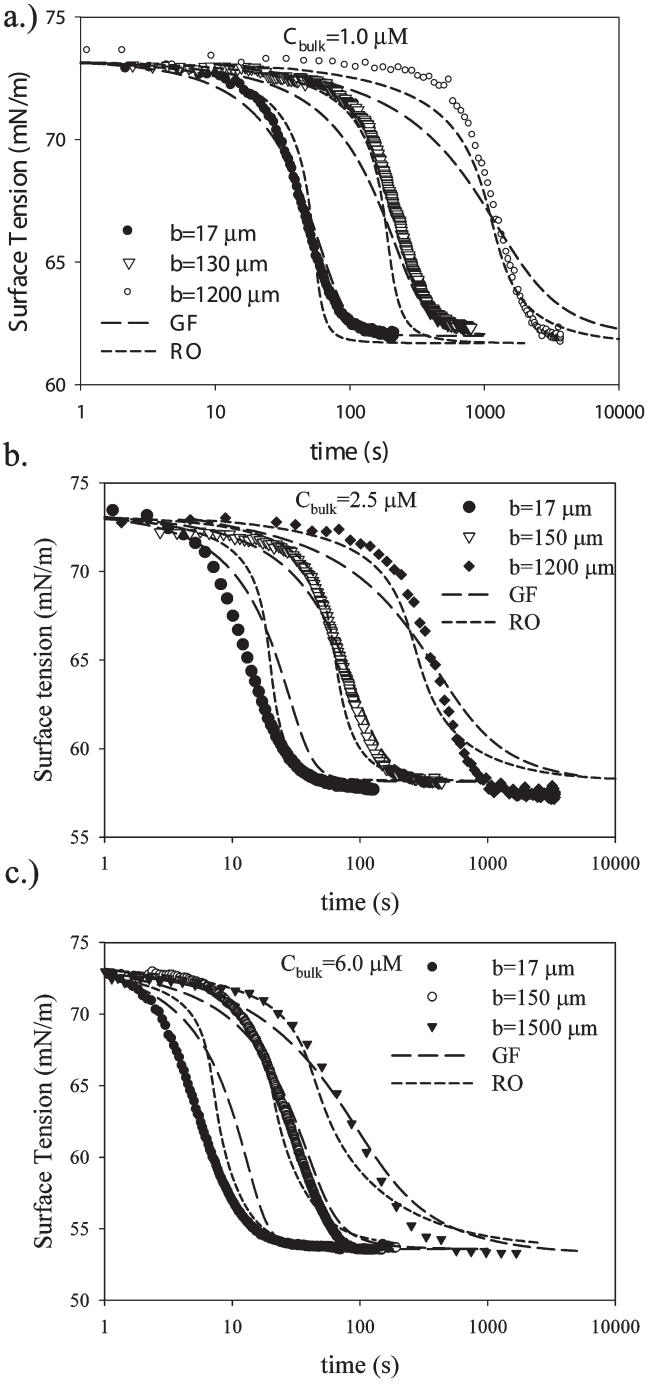



**Figure 9.** Experimental surface tension as a function of time for selected bubble radii compared with diffusion-limited model predictions using the molecular diffusion coefficient, $D = 3.8 \times 10^{-10}$ m$^2$/s, and the generalized Frumkin isotherm. PB refers to pendant bubble experiments.

Figure 6 compares three expansion and compression experiments for different concentrations and bubble radii with dynamics predicted using the GF isotherm. From this figure it is evident that the GF isotherm captures the initial surface tension value that

(36) Nilsson, P. G.; Wennerstrom, H.; Lindman, B. Structure of Micellar Solutions of Non-Ionic Surfactants - Nuclear Magnetic-Resonance Self-Diffusion and Proton Relaxation Studies of Poly(Ethylene Oxide) Alkyl Ethers. *J. Phys. Chem.* **1983**, *87* (8), 1377−1385.

(37) Ferri, J. K.; Stebe, K. J. Soluble surfactants undergoing surface phase transitions: A Maxwell construction and the dynamic surface tension. *J. Colloid Interface Sci.* **1999**, *209* (1), 1−9.

results from compressing or expanding the interface. This is a similar analysis, although not as rigorous, as that previously performed to test the equation of state using the pendant drop apparatus.[6,28,29] In addition, using the molecular value of the diffusion coefficient, the model captures the re-equilibration dynamics quantitatively for different radii and concentrations.

This result is important for two reasons. First, the agreement between model and experiments indicates that diffusion-limited transport can correctly predict the dynamics from expansion and compression experiments using a molecular diffusion coefficient for several different radii and concentrations (i.e., kinetic exchange at the interface is much faster than transport via diffusion). Second, since an expansion experiment only requires a small number of surfactants from the bulk to re-establish equilibrium, diffusion from the bulk is less important than for initially clean interfaces. This is most important for small radii, which have a smaller area to volume ratio and thus more surfactants adjacent to the interface than for larger radii. The fact that the diffusion-limited model works well is further evidence that there is no measurable kinetic barrier in the surfactant transport of $C_{12}E_8$.

## Conclusions

We have developed a new experimental apparatus, a microtensiometer, capable of measuring the dynamic surface tension for bubble radii between 15 and 150 $\mu$m. The technique uses micrometer-scale bubbles pinned at the end of a drawn glass capillary to measure the dynamic surface tension *in situ*. The use of small bubbles reduces the time required to reach equilibrium by an order of magnitude and requires smaller solution volumes than traditional methods involving millimeter-scale bubbles. Smaller bubble radii also allow the pressure to be directly measured using off-the-shelf pressure transducers. We have validated the proper operation of the device using a well-studied nonionic surfactant, $C_{12}E_8$.

Advantages of the microtensiometer apparatus include faster equilibrium dynamics and smaller sample volumes. Since the pendant drop apparatus works at near planar interface conditions, the time to reach equilibrium is the longest possible. However, by decreasing the radius (cf. eq 1), the time scale becomes significantly faster. This will be extremely useful for large molecules such as proteins and other polymers, which have small diffusion coefficients. Solutions of large molecules that reach equilibrium after days with current techniques may now be measured in minutes or hours. This apparatus has the added benefit measuring the effect of curvature on dynamic surface tension, which from Figure 8 has the added advantage of more readily probing the kinetic limited regime than other methods relying on macroscopic radii. In addition, because of the scaling of the depletion depth with both the interface radius and the interface area, the volume of surfactant needed to measure surfactant dynamics in the absence of depletion effects is significantly reduced. Depending on surfactant concentration, the volume of sample required could range from 1 to 2 mL compared with the 15−25 mL required for pendant drop measurements. Finally, the microtensiometer through the use of a high-speed camera (not presented here) has the potential ability to measure subsecond transport dynamics due to the short time required to form the interface.

Through a scaling analysis, the curvature-dependent clean interface dynamics of $C_{12}E_8$ are shown to be diffusion limited for all bubble radii and surfactant concentrations probed. This eliminates the need to include kinetic barriers and reduces the parameter space needed to model the dynamics. More specifically, it reduces the problem of modeling the surfactant transport dynamics to the determination of two unknowns: an isotherm and a diffusion coefficient. Using the molecular value of the self-diffusion coefficient measured independently, we can rigorously test the isotherm using dynamic surface tension measurements. In the case of initially clean interfaces, we show that the GF isotherm cannot accurately capture the dynamics for different radii and concentrations for $C_{12}E_8$. This is possibly due to a gas-to-liquid phase transition occurring at the interface at low surface concentration that cannot be described with a single isotherm. However, using compression−expansion experiments, the phase transition is avoided. We have found that the generalized Frumkin model quantitatively predicts the compression−expansion experiments. This analysis suggests that compression−expansion experiments are a more appropriate test of an isotherm governing surfactant transport in the fully condensed phase.

**Acknowledgment.** The authors thank Professor Kathleen Stebe and Professor Charles Maldarelli for helpful discussions. N. J. Alvarez thanks G. Labrosse, University of Paris, Orsay, and A. K. Uguz, Bogazici University, for helpful discussions concerning spectral methods and the National Science Foundation graduate student fellowship for financial support. Partial funding was provided by NSF Grant CBET-0730727.